

DATA ANALYTICS PORTFOLIO PROJECT

NexaMart Customer Profitability & Revenue Intelligence

End-to-End Retail Business Analytics Project

Python

SQL

Power BI

Excel

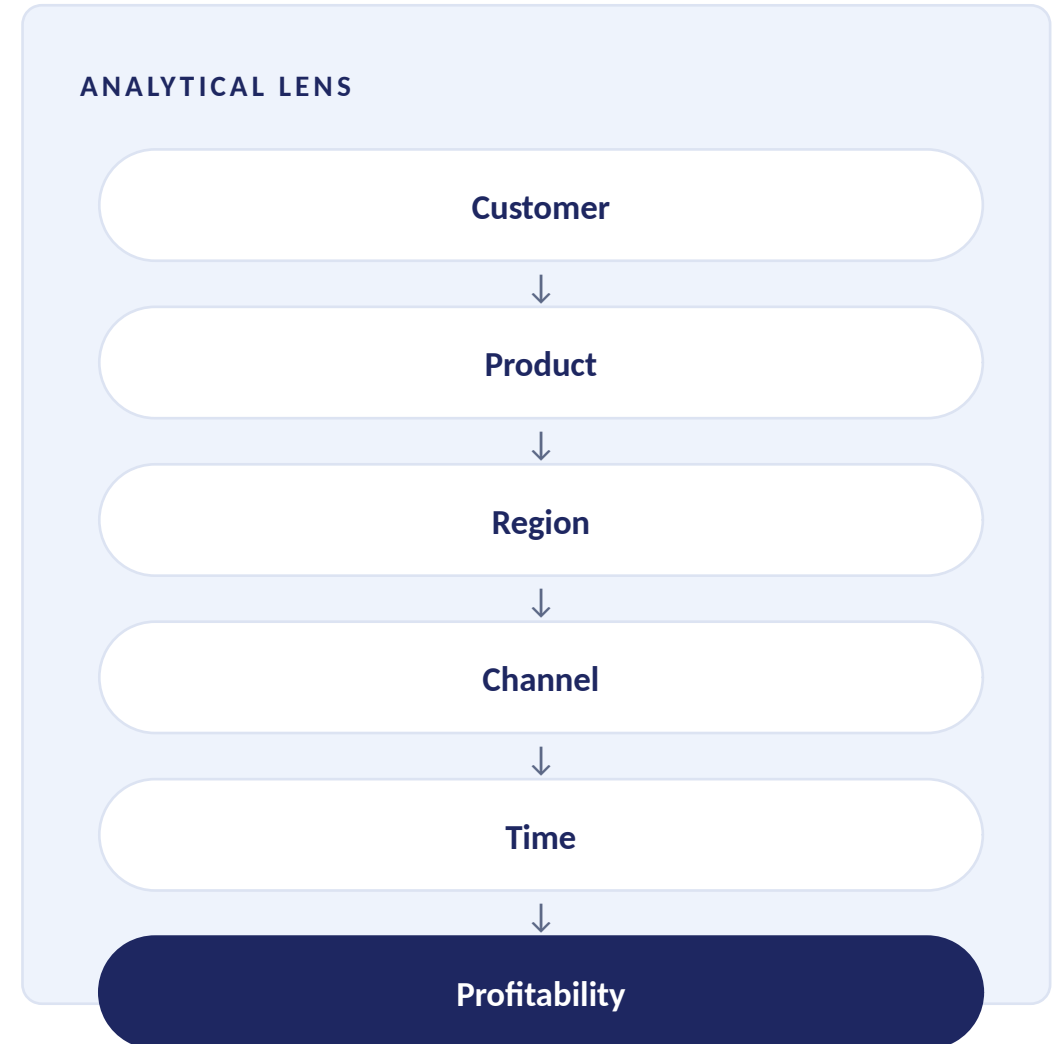
Revenue Scale Does Not Guarantee Profit

NexaMart is a multi-category retail business operating across multiple regions and two sales channels. As the business has grown, revenue and profit have not moved in lockstep — raising a core management question.

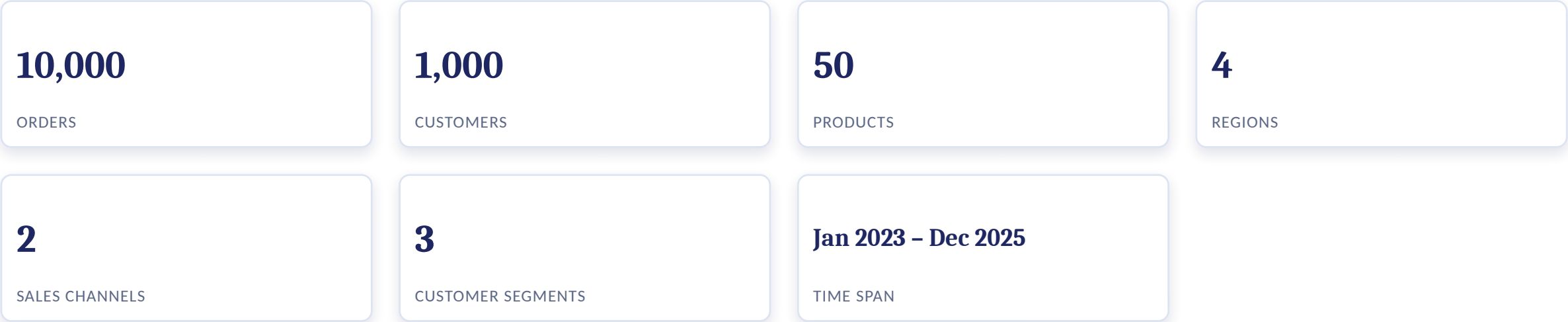
“Where is NexaMart generating revenue, and where is it actually creating profitable value?”

This analysis was built to answer five questions:

- 1 Which customers are most valuable?
- 2 Which products / categories are most profitable?
- 3 Which regions perform best?
- 4 Which channel is more efficient?
- 5 How do discounts affect profitability?



Project Scope at a Glance

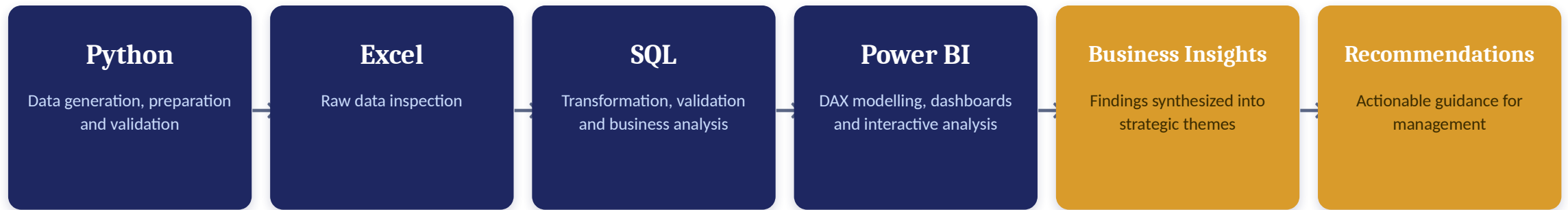


Synthetic transactional retail dataset — modeled to reflect realistic multi-category retail behaviour.

ANALYTICAL DIMENSIONS

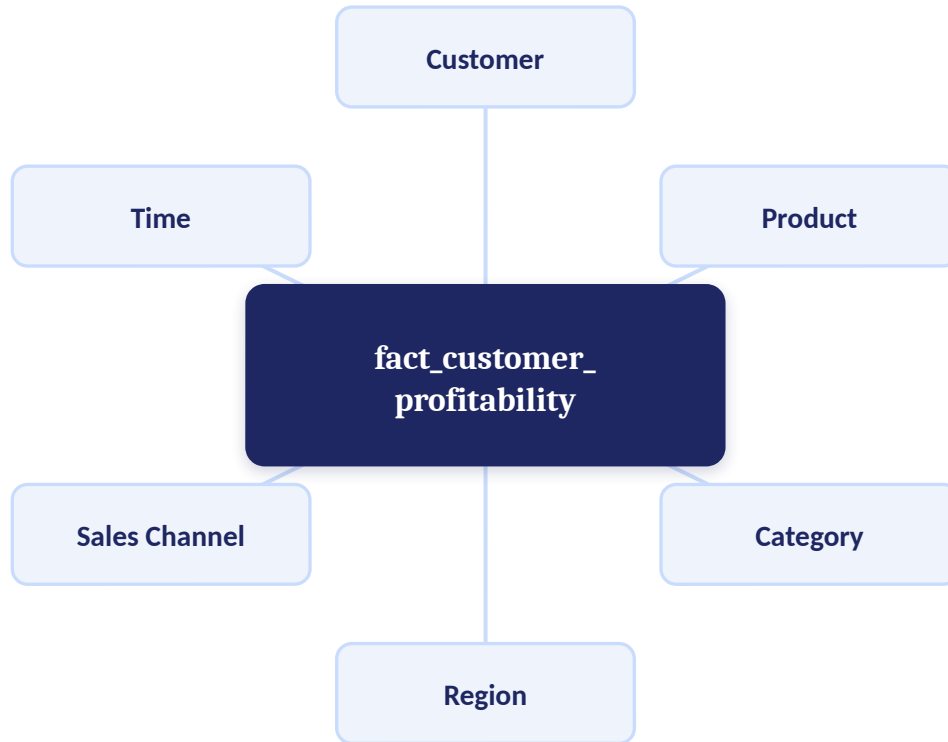


End-to-End Analytical Workflow



Four tools, one narrative: raw transactions become validated data, then modeled metrics, then decisions.

Data Model & Analytical Architecture



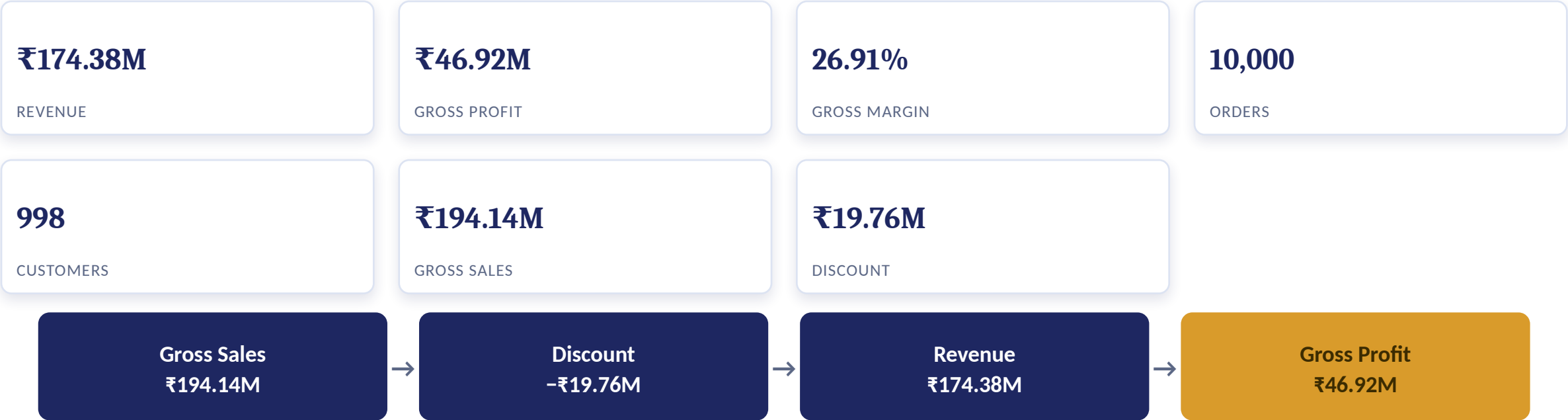
CORE METRICS

- Revenue
- COGS
- Gross Profit
- Gross Margin
- Discount

*Grain: one row per transaction / order.
Order_ID is the transaction identifier.*

A single fact table at order grain, joined to six dimensions — enabling profitability analysis by any combination of customer, product, region, channel and time.

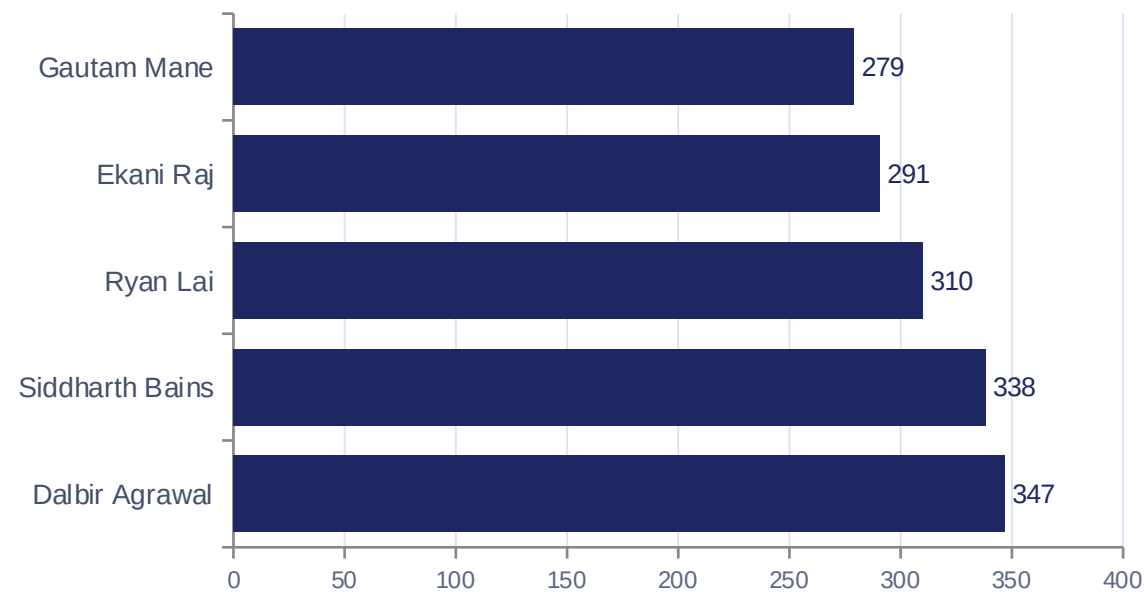
Executive Business Performance



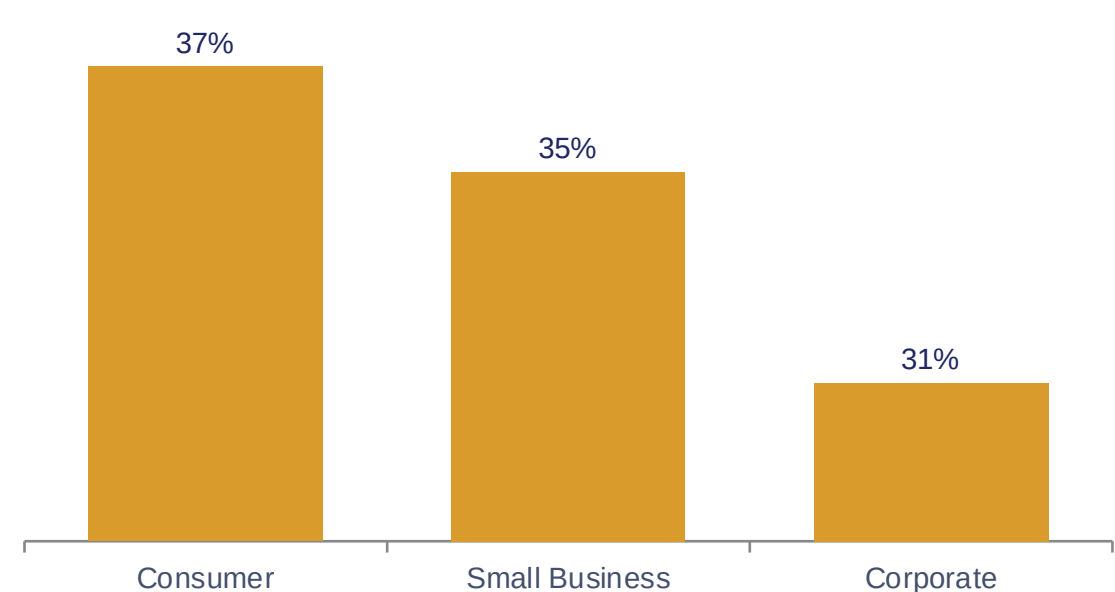
EXECUTIVE INSIGHT NexaMart has a strong revenue and gross-profit base, but profitability varies significantly across customers, products, regions and channels.

Customer & Segment Profitability

TOP CUSTOMERS BY GROSS PROFIT



SEGMENT GROSS MARGIN

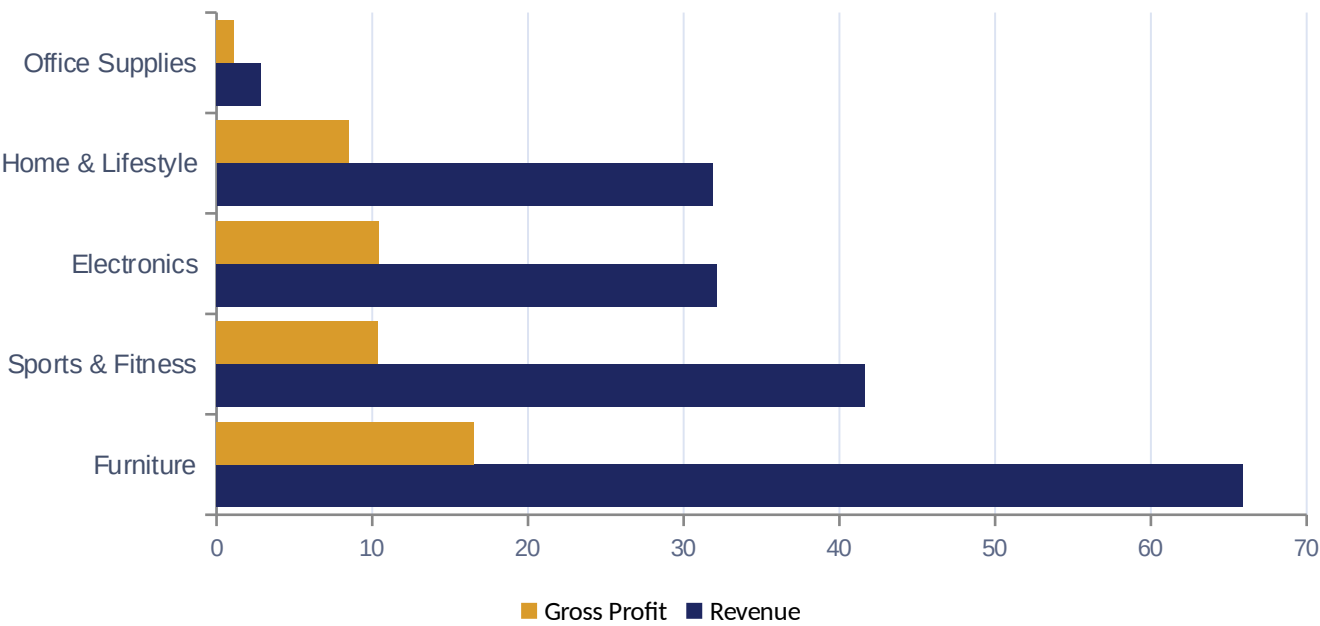


Top gross-profit customers are concentrated in the Corporate segment.

INSIGHT Corporate customers generate strong absolute customer-level profit, while Consumer customers demonstrate higher margin efficiency — customer value and margin efficiency are two different lenses.

Product & Category Performance

CATEGORY: REVENUE vs GROSS PROFIT (₹M)



TOP PRODUCTS BY GROSS PROFIT

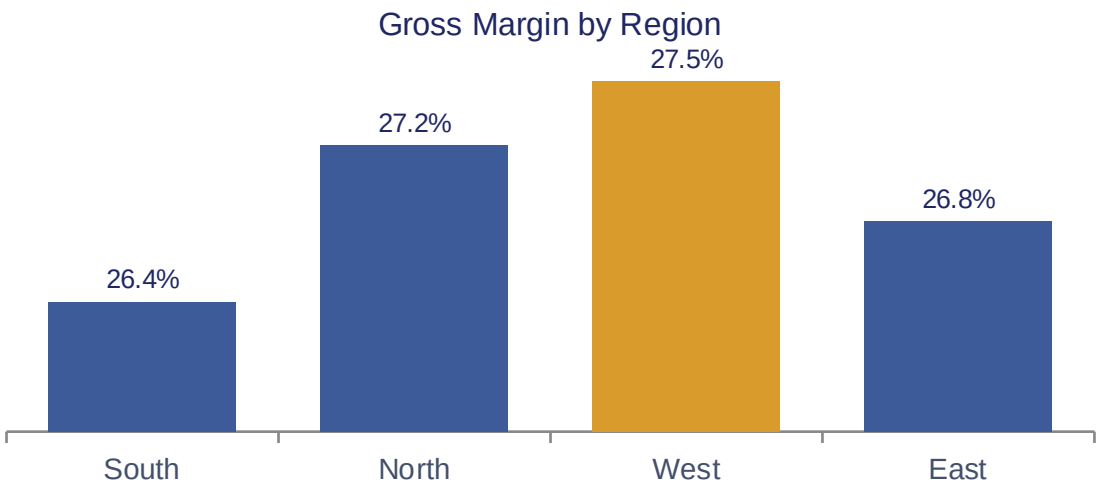
Exercise Bike Margin 19.80%	₹4.97M GP
Standing Desk Margin 23.48%	₹3.59M GP
Exec. Office Chair	₹2.58M GP

Furniture = largest scale · Electronics = strong scale + margin combo · Office Supplies = highest margin, smallest scale (38.74%)

KEY MESSAGE Revenue leadership and margin leadership are not the same.

Regional Performance

Region	Revenue (₹M)	Gross Profit (₹M)	Margin	Orders
South	69.83	18.45	26.43%	3,821
North	44.07	11.98	27.19%	2,608
West	38.50	10.59	27.50%	2,318
East	21.98	5.89	26.82%	1,253



INSIGHT South provides scale; West provides a margin-efficiency benchmark.

South = largest revenue and gross-profit contributor. **West** = highest gross margin at 27.50%.

Sales Channel Performance

STORE

Leads in Scale

Revenue	₹92.33M
Gross Profit	₹24.36M
Gross Margin	26.39%
Orders	4,564

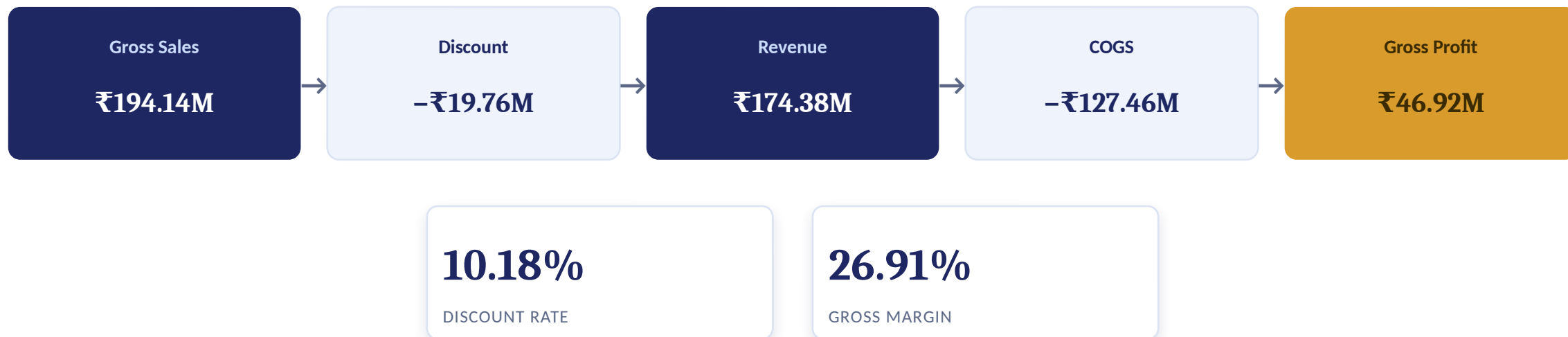
ONLINE

Leads in Efficiency

Revenue	₹82.06M
Gross Profit	₹22.56M
Gross Margin	27.49%
Orders	5,436

INSIGHT Store leads in scale, while Online leads in efficiency and order volume — Online carries roughly 1.10 percentage points higher gross margin.

Discount & Profitability



INSIGHT Discounting is a significant profitability lever and should be managed based on incremental value rather than sales volume alone.

Key Business Insights

1 CUSTOMER

Corporate customers dominate top individual profit contributors.

2 MARGIN

Consumer has the highest segment margin at 37%.

3 PRODUCT

Exercise Bike is the leading identified product by gross profit.

4 CATEGORY

Furniture leads in absolute revenue and gross profit.

5 REGION

South leads in scale; West leads in margin efficiency.

6 CHANNEL

Store leads in absolute profit; Online leads in margin and order volume.

Management Recommendations

1 Protect and retain high-profit Corporate customers.

2 Investigate whether Corporate margins can be improved via pricing, discount management or product mix.

3 Use West as a benchmark for regional margin efficiency.

4 Prioritize Electronics — it combines significant revenue with ₹10.44M gross profit at a 32.46% margin.

5 Evaluate scaling Office Supplies while protecting its 38.74% margin.

6 Expand the Online channel intelligently — higher order volume and higher gross margin.

7 Shift from broad discounting toward targeted discounting based on profitability.

CONCLUSION

From Transaction Data to Profitability Intelligence



Revenue scale, customer value, and margin efficiency are different dimensions of business performance. A profitable growth strategy should optimize all three — not focus on revenue alone.

TECHNOLOGY: Python | SQL | Power BI | Excel

Validation: data checks · financial calc checks · SQL validation · Power BI validation · filter/slicer testing · cross-source comparison